

COMBIENBAR, Australia

WMO: 949140

Lat: 37.3417S

Lon: 149.0228E

Elev: 641

StdP: 93.86

Time Zone: 10.00 (AUS)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	2.3	3.1	-5.8	2.5	11.0	-3.3	3.1	9.6	7.5	8.3	6.5	8.3	2.2	230	0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.3	32.3	18.5	29.9	18.0	27.8	17.6	20.6	27.9	19.7	26.6	18.8	25.3	2.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	14.4	23.2	17.5	13.5	22.0	16.5	12.7	21.2	62.3	27.9	59.0	26.5	55.9	25.3	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6.4	5.2	4.3	0.7	36.5	0.6	2.6	0.3	38.3	-0.1	39.8	-0.4	41.2	-0.9	43.1
			-0.2	22.6	0.5	0.8	-0.6	23.2	-0.9	23.7	-1.2	24.2	-1.6	24.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.1	18.9	18.3	16.4	13.4	10.8	8.6	7.9	8.6	10.6	12.6	14.8	16.6
	DBStd	5.26	4.76	4.31	4.02	3.38	3.01	2.37	2.24	2.76	3.38	4.09	4.50	4.60
	HDD10.0	282	0	0	1	7	26	56	73	61	32	18	6	2
	HDD18.3	2127	51	51	87	152	234	293	324	302	234	185	125	89
	CDD10.0	1412	277	233	201	109	50	13	7	18	49	97	151	207
	CDD18.3	215	69	50	29	4	0	0	0	0	1	6	20	37
	CDH23.3	2029	726	441	204	18	0	0	0	0	4	54	190	392
	CDH26.7	712	303	148	56	2	0	0	0	0	1	9	54	140
(14) Wind	WSAvg	1.3	1.3	1.1	1.0	1.0	1.3	1.3	1.5	1.5	1.5	1.5	1.4	1.3
(15) Precipitation	PrecAvg	920	69	83	64	80	65	104	72	71	67	81	88	77
	PrecMax	1158	156	216	234	278	195	326	153	134	151	201	186	169
	PrecMin	651	18	16	6	8	21	11	9	28	20	20	20	16
	PrecStd	154	37	50	52	64	38	85	47	33	32	40	42	39
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.4	33.9	31.2	26.0	20.5	16.6	16.2	19.1	23.4	27.9	31.4	34.1
		MCWB	20.0	19.3	18.4	16.1	12.8	11.2	9.2	11.3	12.9	14.9	17.2	18.3
	2%	DB	32.7	30.6	28.0	22.6	18.4	14.7	14.1	16.6	20.7	24.7	28.0	30.4
		MCWB	19.3	18.4	17.2	15.0	12.1	9.7	8.6	10.0	12.0	14.2	16.2	17.6
	5%	DB	30.0	28.1	25.4	20.7	16.8	13.4	12.6	14.7	18.7	22.3	25.3	27.7
		MCWB	18.7	18.1	16.8	14.1	11.4	8.9	7.9	9.0	11.4	13.4	16.0	17.4
	10%	DB	27.4	26.1	23.2	18.9	15.3	12.1	11.4	13.2	16.6	20.0	22.9	25.1
		MCWB	18.0	18.0	16.5	13.3	10.6	8.2	7.1	8.2	10.5	12.7	15.4	16.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.2	21.6	20.4	17.3	14.5	12.2	10.7	12.3	14.5	16.9	19.4	20.7
		MCDB	29.9	27.9	26.5	23.4	17.6	14.9	14.2	17.2	20.7	23.0	25.9	27.6
	2%	WB	20.7	20.4	19.0	15.8	13.2	10.7	9.4	10.7	13.1	15.5	17.8	19.3
		MCDB	29.1	26.8	24.5	20.7	16.8	13.8	12.7	15.0	18.8	21.6	24.6	26.7
	5%	WB	19.7	19.4	17.9	14.7	12.2	9.7	8.5	9.6	12.0	14.4	16.7	18.1
		MCDB	27.3	25.9	23.2	19.1	15.7	12.3	11.4	13.8	17.2	20.5	23.1	25.3
	10%	WB	18.7	18.4	16.9	13.8	11.2	8.8	7.8	8.6	10.9	13.3	15.7	17.1
		MCDB	25.7	24.3	22.1	17.8	14.6	11.3	10.5	12.4	15.8	19.0	21.9	23.8
(35) Mean Daily Temperature Range	5% DB	MDBR	11.3	10.3	9.0	7.5	5.7	5.0	5.3	6.5	8.3	9.8	10.5	11.0
		MCDBR	15.5	14.2	12.4	10.1	7.5	6.8	7.2	8.8	11.5	13.5	14.4	15.2
		MCWBR	6.7	6.2	5.8	5.3	4.8	4.6	4.8	5.4	6.1	6.4	6.7	6.8
	5% WB	MCDBR	13.7	12.2	10.6	8.9	6.9	6.1	6.4	8.3	10.4	12.0	12.8	13.1
		MCWBR	6.9	6.3	5.8	5.2	4.7	4.5	4.6	5.4	6.1	6.5	6.8	7.0
(40) Clear-Sky Solar Irradiance	taub	0.392	0.357	0.320	0.306	0.288	0.278	0.270	0.280	0.302	0.319	0.328	0.341	(40)
	taud	2.350	2.488	2.589	2.604	2.624	2.653	2.687	2.643	2.572	2.543	2.550	2.517	(41)
	Ebn at Noon	941	952	953	906	868	852	885	924	956	981	999	997	(42)
	Edh at Noon	131	110	92	79	67	60	62	74	91	103	107	112	(43)
(44) All-Sky Solar Radiation	RadAvg	6.56	5.62	4.73	3.51	2.56	2.01	2.32	3.17	4.36	5.47	6.13	6.71	(44)
	RadStd	0.47	0.42	0.37	0.27	0.13	0.15	0.16	0.19	0.31	0.33	0.45	0.48	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46) Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47) Regional (25 neighbors)	+0.51	+0.35	N/A	N/A	+0.65	+0.62	N/A	-108	+172	+68
	+0.35	N/A	N/A	N/A	N/A	N/A	N/A	-87	+126	+47

Nomenclature: See separate page